

Homework 1

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Problem 1. Let X be an affine variety. Show that the coordinate ring $A(X)$ is a field if and only if X is a single point.

Proof. ($\implies$) Let $X = \{p\} \subseteq \mathbb{A}_k^n$ be a point, and write $p = (p_1, \dots, p_n)$. Note that p is precisely the vanishing locus of the ideal $J = (x_1 - p_1, \dots, x_n - p_n)$. Note that this is clearly a maximal ideal, since $k[x_1, \dots, x_n]/J \cong k$ is a field. By Hilbert's Nullstellensatz, $A(X) = k[x_1, \dots, x_n]/I(V(J)) \cong k[x_1, \dots, x_n]/\sqrt{J}$. Since all maximal ideals are radical, $\sqrt{J} = J$. Hence, $A(X) \cong k$.

($\impliedby$) Now let X be an affine variety such that $A(X)$ is a field. Then $I(X)$ is necessarily a maximal ideal. By the above argument, any point $p \in X$ yields an ideal $I(p) = (x_1 - p_1, \dots, x_n - p_n)$. Since Hilbert's Nullstellensatz is inclusion-reversing and $\{p\} \subseteq X$, we have $I(X) \subseteq I(p)$. Since $I(X)$ is maximal, we have $I(X) = I(p)$. Since $V(I(X)) = X$, we have $X = V(I(p)) = \{p\}$ by the Nullstellensatz, hence X is a point. ■

Problem 2. Determine the radical of the ideal $J = \langle x_1^3 - x_2^6, x_1x_2 - x_2^3 \rangle \subseteq \mathbb{C}[x_1, x_2]$. (Hint: Hilbert's Nullstellensatz may be useful here.)

Proof. Sketch of the idea: by Hilbert's Nullstellensatz, $\sqrt{J} = I(V(J))$. The vanishing locus of this ideal should be computable. The radical is then just all functions that vanish on this locus. The first generator probably give something like $x_1 - x_2^2$, and idk what the right is.

Note that $x_1x_2 - x_2^3 = x_2(x_1 - x_2^2)$. ■

Problem 3. Let $X \subseteq \mathbb{A}_k^3$ be the union of the three coordinate axes. Compute generators for the ideal $I(X)$, and show that $I(X)$ cannot be generated by fewer than 3 elements.

Proof. A useful result might be Lemma 1.4 in Galthmann's notes, which says that $V(S_1) \cup V(S_2) = V(S_1S_2)$. This implies $X = V(xyz)$, and the Nullstellensatz says $I(X) = \sqrt{\langle xyz \rangle}$.

Galthmann Lemma 1.4 says that $V(S_1) \cup V(S_2) = V(S_1S_2)$. Since the x_i -axis is precisely $V(x_i)$, we see that $X = V(xyz)$. Thus $f \in I(X)$ if and only if $xyz = 0$ implies $f(x, y, z) = 0$.

Since k is a field, $k[x, y, z]$ is an integral domain, hence it has no zero divisors. Thus $xyz = 0$ implies at least one of x, y, z is 0. We claim $I(X) = \langle x, y, z \rangle$. It suffices to show that all terms of some $f \in I(X)$ are of the form $\lambda x^n y^m z^\ell$, since such a term is necessarily in $\langle x, y, z \rangle$. Write

$$f(x, y, z) = \sum_{n,m,\ell \geq 0} f_{n,m,\ell} x^n y^m z^\ell,$$

where $f_{n,m,\ell} \in k$ are the coefficients of f .

Recall that ideals of a ring R are precisely R -submodules of R . Hence, to show $I(X)$ cannot be generated by less than 3 elements, it suffices to show that $\{x, y, z\}$ is a basis for the free module $I(X)$. Since we have shown that they span the space, it suffices to show $k[x, y, z]$ -linear independence. Since there are no relations between x, y, z , this is so trivial that it's difficult to come up with a proof which rigorously expresses this fact. I came up with the following. Recall that

$$k[x] \otimes k[y] \otimes k[z] \cong k[x, y, z]$$

via the identification $x \otimes y \otimes z \mapsto xyz$. Hence we can equivalently show that the elements $x \otimes 1 \otimes 1$, $1 \otimes y \otimes 1$, and $1 \otimes 1 \otimes z$ are linearly independent. ■

Problem 4. Let $Y \subseteq \mathbb{A}_k^n$ be an affine variety, and let $\pi : k[x_1, \dots, x_n] \rightarrow A(Y)$ be the quotient map.

1. Show that $V_Y(J) = V(\pi^{-1}(J))$ for every ideal $J \subseteq A(Y)$.
2. Show that $\pi^{-1}(I_Y(X)) = I(X)$ for every affine subvariety $X \subseteq Y$.
3. Use (a) and (b) to deduce the interesting part of the relative Nullstellensatz $I_Y(V_Y(J)) \subseteq \sqrt{J}$ for every ideal $J \subseteq A(Y)$ from the corresponding absolute statement $I(V(J)) \subseteq \sqrt{J}$ for every ideal $J \subseteq k[x_1, \dots, x_n]$. In particular, conclude that there is an inclusion-reversing bijection between affine subvarieties of Y and radical ideals of $A(Y)$.